

Refining the Computational Analysis of Interactive Fiction: A Hybrid Framework using Bag-of-Paths Formalism and LLMs

Guillaume Guex guillaume.guex@unil.ch

University of Lausanne

Introduction

Interactive Fiction and Gamebook

An **interactive fiction** is a form of narrative in which the reader's or player's choices influence the sequence of events, the accessible content, or the story's outcome.

A **gamebook** is a book-based form of interactive fiction, usually divided into numbered sections connected by choices, rules, and sometimes game mechanics such as combat, inventory, or chance.

112

Suddenly, the large rock you are hiding behind is rolled aside and you are faced by two snarling Giaks intent on your death. The cave mouth is a narrow entrance and you can only fight the Giaks one at a time.



Giak 1: COMBAT SKILL 13 ENDURANCE 10

Giak 2: COMBAT SKILL 12 ENDURANCE 10

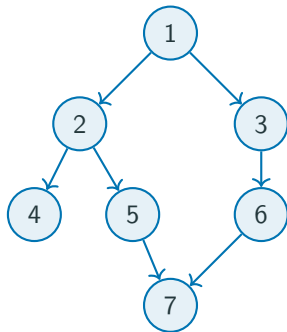
If you win, you may explore the cave further by **turning to 33**.

Or you may leave and descend the hill. **Turn to 248**.

The classical representation: a graph

Interactive fiction is traditionally modeled as a **directed graph**.

- **Nodes** represent *paragraphs*, i.e. numbered narrative sections.
- **Edges** represent *transitions*, i.e. the choices available to the player.



The **Main limitation** of this representation:

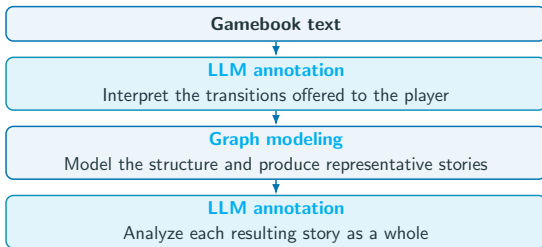
- The **structure** or the **topology** of the work can be studied.
- The text itself, therefore the **narrative meaning**, is no longer taken into account.

A hybrid solution: graphs and LLMs

Large Language Models (LLMs) make it possible to reintroduce the semantic content discarded by graph modeling.

Our approach: both methods

- The **graph** models the **structure** of the work
- The **LLM** interprets the **narrative content**



Research question

How can graph modeling and local LLM annotation be combined to study both the structural possibilities and the narrative experiences produced by a gamebook?

Methodological choice: local LLM

Methodological choice

- We are going to use exclusively [Qwen3.6-27B](#), executed locally on university infrastructure.
- The local model is not treated as an autonomous literary critic. It is used as a [controlled annotation instrument](#) whose decisions remain open to inspection.

Methodological reasons

- No dependency on a proprietary API
- Prompts and inference settings are documented
- Outputs can be stored, inspected and reproduced
- Source texts remain on local infrastructure
- Lower and predictable financial cost

Why is it sufficient?

- The task is a constrained annotation task
- Categories are defined by an explicit codebook
- Outputs follow a restricted JSON schema
- Results are checked against human annotations
- Frontier-level generative capabilities are overkill

Corpus: Lone Wolf and Project Aon

The studied corpus is the **Lone Wolf series** created by Joe Dever.

The **Project Aon** (www.projectaon.org) provides authorized digital editions of the books. It is already structured with html.

There are **several mechanics** in these gamebooks:

- Branching choices and forced transitions
- Random events
- Player profile with stats and special powers
- Combat with possible evasion
- Endurance (HP) management
- Items and inventory management
- Puzzles and hidden links

Scope of this presentation

A complete representation of every mechanic would make the model intractable. We use **explicit simplifications** and focus exclusively on **Book 1: Flight from the Dark**.

Modeling scope: from L0 to L3

The goal is to build a tractable **Markovian model**: for a fixed player profile, the next transition depends only on the current node, not on the complete play history.

Level	Information	Mechanics	Decision
L0	Topology	Paragraphs, links, forced transitions, Death and Win outcomes	Included
L1	Player agency	Choices annotated along the risk, morality and action axes	Included
L2	Uncertainty	Random events, combat, evasion, Kai Disciplines (special powers) and conditions	Simplified
L3	Persistent state	Endurance, healing, inventory, gold, equipment and meals	Not simulated

L2 simplifications:

- One global combat-win probability
- One global escape probability
- Kai Disciplines availability fixed at 0.5
- One generic `has_condition` probability

Outline of the presentation

1. Paragraph parsing and transition annotation

A local LLM reads the choices offered at each paragraph and outputs:

- Transition types
- Game mechanics
- Risk, morality and action labels

2. Graph modeling and Bag-of-Paths formalism

The annotations are compiled into weighted graphs for different player profiles, and various objects are computed:

- Bag-of-Paths indices
- Local and global measures
- Representative trajectories

3. Trajectory annotation

A local LLM reads the stories associated with selected trajectories and produces:

- Whole-story annotations
- Profile manifestation
- Pairwise comparisons

Paragraph parsing and transition annotation

Objective: from paragraphs to structured data

Objective

Transform each gamebook paragraph into an **structured representation**, separating **narrative units** from the **transitions** offered to the player.

Project Aon page

112

Suddenly, the large rock you are hiding behind is rolled aside and you are faced by two snarling Giaks intent on your death. The cave mouth is a narrow entrance and you can only fight the Giaks one at a time.



Giak 1: COMBAT SKILL 13 ENDURANCE 10
Giak 2: COMBAT SKILL 12 ENDURANCE 10

If you win, you may explore the cave further by **turning to 33**.

Or you may leave and descend the hill. **Turn to 248**.

Paragraph 112



Parsing

+

LLM

LW01_nodes.csv

node_id	text_content	absorbing_status	enemies
112	Suddenly, the large rock...	potential_death	Giak 1; Giak 2

LW01_edges.csv

source_id	target_id	edge_text	transition_type	realisation_value
112	33	If you win, explore the cave...	explicit_choice	If you win
112	248	Leave and descend the hill...	explicit_choice	If you win

source_id	target_id	semantic_risk	semantic_morality	semantic_action	warnings
112	33	reckless	neutral	tactical	-
112	248	cautious	selfish	physical	-

Rule-based parsing versus LLM-based annotations

Rule-based parsing (on html)

Can process all fields in the **node table**, and **some fields in the edges tables** (source_id, target_id, and edge_text, used for validation).

LLM-based annotations

All remaining fields in the **edge table**, in particular:

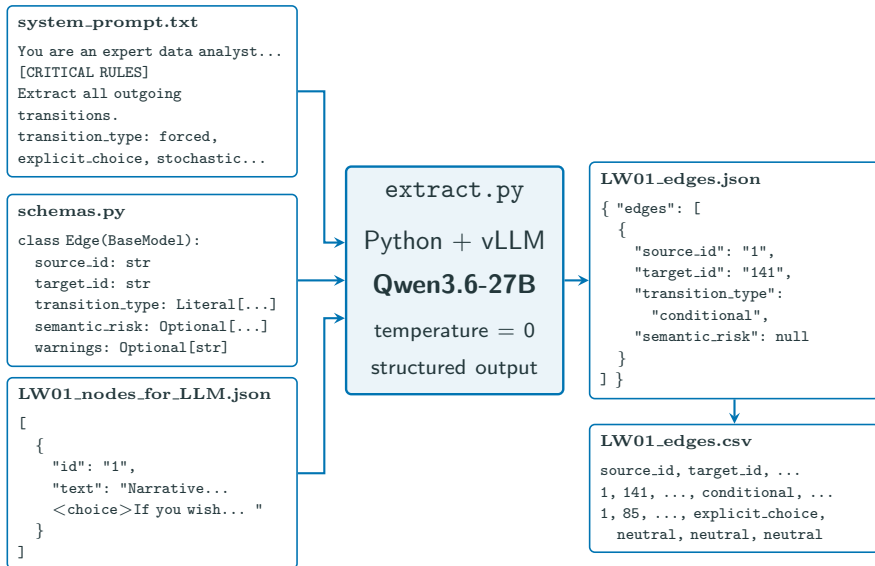
- Transition mechanism: transition_type, realisation_value;
- Meaning of player choices: semantic_risk, semantic_morality, semantic_action;
- Ambiguities requiring attention: warnings.

Methodological choice

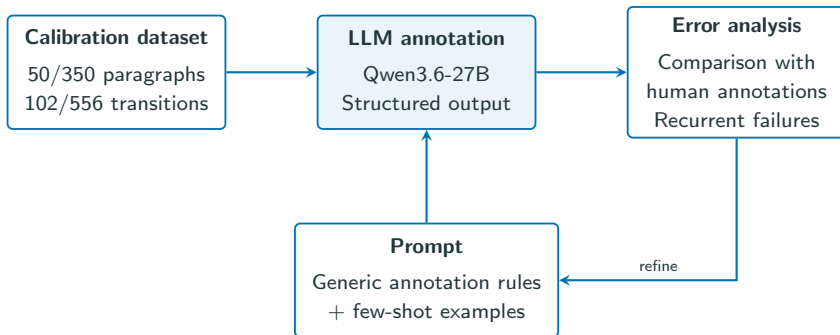
Each explicit player choice is annotated according to **3 separate semantic axes**:

- **Risk**: exposure to danger (cautious, neutral, reckless)
- **Morality**: treatment of others (selfish, neutral, noble)
- **Action**: mode of intervention (physical, neutral, tactical)

From prompt to structured annotation



Iterative prompt refinement



- Errors lead to revisions of **general annotation rules**, not paragraph-specific corrections.
- This iterative process is a **calibration**, not an independent evaluation.
- No held-out validation set was used because of limited time; generalization to another book is therefore not measured.

Annotation results

Calibration: On 50 paragraphs and 102 transitions, the final prompt produced no missing or invented transition and no transition-type disagreement. Only 3 (of 177 annotations) semantic disagreements: two for risk and one for action.

Paragraph statistics

Total	350
Terminal	16 deaths / 1 victory
One outgoing transition	157
Two outgoing transitions	134
Three or more outgoing transitions	42
Paragraphs with enemies	29

Transition statistics

Transition type	N	%
Explicit choice	291	52.3
Forced	157	28.2
Conditional	66	11.9
Stochastic	42	7.6
Total	556	100.0

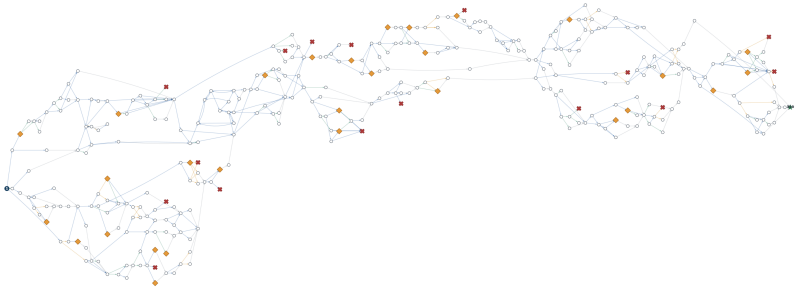
Axis	First pole	Neutral	Opposite pole
Risk	cautious: 95 (32.6%)	84 (28.9%)	reckless: 112 (38.5%)
Morality	selfish: 18 (6.2%)	262 (90.0%)	noble: 11 (3.8%)
Action	physical: 101 (34.7%)	126 (43.3%)	tactical: 64 (22.0%)

Warnings	53 (9.5%)
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Result of paragraph annotation

LW01 — annotated topology

— Explicit choice — forced — Stochastic — Conditional — Combat paragraph — Death ending — Victory ending



330 paragraph nodes and 536 directed transitions. Colors show phase 2 annotations; no profile or probability is applied. Layout coordinates: Project Aem.

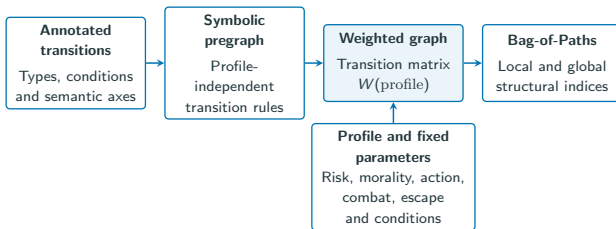
The topology and transition types are now available, but **transition probabilities depend on player profiles** (next section).

Graph construction and Bag-of-Paths analysis

Objective: from annotations to structural measures

The objectives here are to:

- Transform annotations into **weighted directed graphs (i.e. a Markov chain matrix)** which depend on **player profiles**.
- Compute **various indices with the Bag-of-Paths formalism**.



This section addresses three questions:

- How can heterogeneous game mechanics be converted into transition probabilities?
- How does the resulting graph vary across player profiles?
- Which paragraphs and structural properties are most important across all possible trajectories?

One Markov chain per player profile

The matrix $W^{(p)}$ describes the behavior of one fixed player profile. Once the profile and the experimental parameters are fixed, the next transition depends only on the current paragraph.

Player profile

$$p = (\text{risk}, \text{morality}, \text{action})$$

- Risk: cautious, neutral, reckless
- Morality: selfish, neutral, noble
- Action: physical, neutral, tactical

$$3^3 = 27 \text{ player profiles}$$

Fixed parameters (all profiles)

- Combat victory: $P(\text{win}) = 0.833$
- Combat evasion: $P(\text{escape}) = 0.5$
- Required Kai Discipline available:
 $P(\text{Kai}) = 0.5$
- Item, gold, or Endurance condition satisfied:
 $P(\text{has_condition}) = 0.5$

From profile to probability

Each choice annotation is compared with the profile:

Relation	Affinity
Matching non-neutral values	2
Neutral profile or annotation	1
Opposed non-neutral values	0.5

Affinities are multiplied and normalized among the available outgoing choices.

For a reckless profile facing one reckless and one cautious choice:

$$P(\text{reckless}) = \frac{2}{2+0.5} = 0.8$$

$$P(\text{cautious}) = 0.2.$$

Bag-of-Paths at the Random-Walk limit

For each player profile, $\mathbf{W}^{(p)}$ represents the corresponding Markov chain. For the BoP calculation, absorbing states have no outgoing transitions: $\mathbf{W}^{(p)}$ is therefore **substochastic**, and we can compute:

$$\mathbf{z}^{(p)} = \left(\mathbf{I} - \mathbf{W}^{(p)} \right)^{-1} = \mathbf{I} + \mathbf{W}^{(p)} + \left(\mathbf{W}^{(p)} \right)^2 + \dots$$

$z_{si}^{(p)}$ is the **expected number of visits to paragraph i when starting from paragraph s** . Many indices can be derived from the same fundamental matrix:

Local indices

Computed for individual paragraphs or transitions:

- Probability of visiting a paragraph
- Expected visits and edge flows
- Contribution to mortality
- Impact of a choice on the final outcome
- Sensitivity to the player profile

Global indices

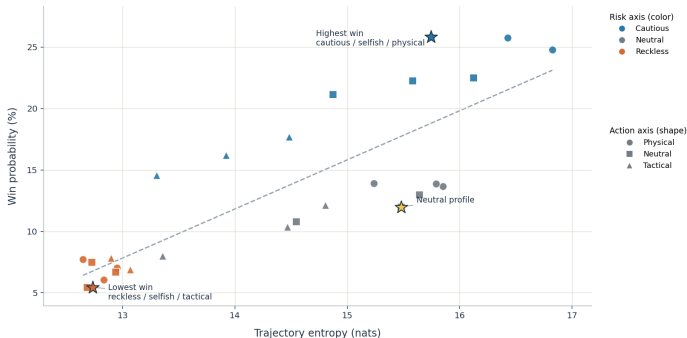
Computed for the complete book and one profile:

- Probabilities of victory and death
- Expected trajectory length
- Total trajectory entropy
- Expected coverage and replayability
- Divergence between player profiles

Player profiles change survival and narrative freedom

LW01 — player profiles change both survival and narrative freedom

Across the 27 configured profiles, win probability and path entropy rise together (descriptive $r = 0.83$).



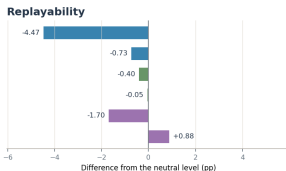
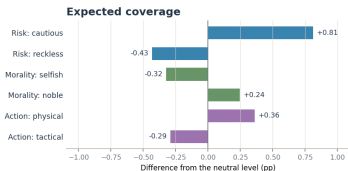
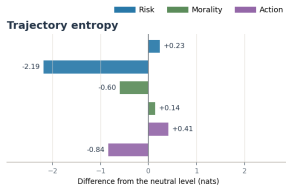
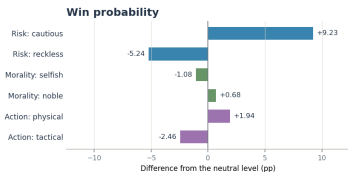
Each point is one profile. The correlation describes the complete factorial design; it is not a population estimate or a causal effect.

Across the **27 configured profiles**, win probability ranges from **5.4% to 25.8%**. Survival and trajectory entropy rise together ($r = 0.83$): profiles that survive longer also access a wider range of paths.

Risk is the dominant behavioral axis

LW01 — risk is the dominant behavioral axis

Marginal effects average over the nine combinations of the other two axes.



Percentage-point differences are used for probabilities. Each comparison uses the neutral level of the same axis as its baseline.

Moving from neutral to cautious choices increases win probability by **9.23 percentage points**; moving to reckless choices reduces it by **5.24 points**. Morality and action have smaller, but non-zero, effects.

Different indices reveal different key paragraphs

LW01 — three local views of the same narrative graph

The Project Aon layout is fixed across panels; only the BoP node metric changes.

● Ordinary ◆ Combat ☒ Death ★ Win ● Start

A Neutral visit probability — the common narrative backbone



B Mortality contribution — where neutral runs end



C Profile sensitivity — where player types diverge



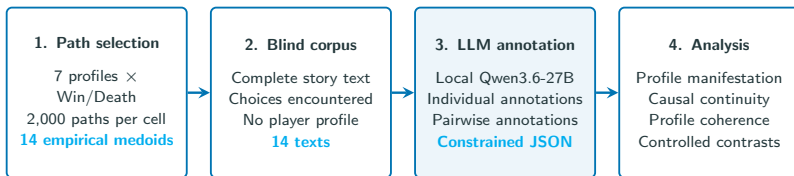
Node size and color encode the panel metric. Faint edges are scaled by neutral expected flow. Labels identify each panel's five largest values plus \$1 and \$350.

The common narrative backbone, the main mortality bottlenecks, and the regions where player profiles diverge **do not coincide**. Local importance therefore depends on the structural question being asked.

Trajectory annotation

Objective: from probabilistic paths to complete stories

This section asks whether the structural differences produced by player profiles become **perceptible differences in complete stories**.

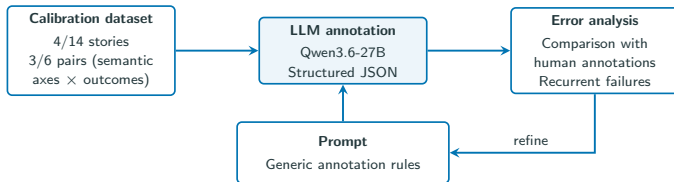


A **medoid** is an observed trajectory that minimizes its average distance to all other sampled trajectories, making it the most central representative path.

The model infers whether the intended profile becomes perceptible from the **story and choices only**. The hidden metadata and BoP measures are reintroduced only for the final comparison.

LLM annotation: calibration and output fields

Same iterative procedure as in paragraph annotation, now applied to **complete trajectories**.



The 3rd prompt was frozen: 7/7 valid outputs, no inadmissible evidence, and 32/44 concordances.

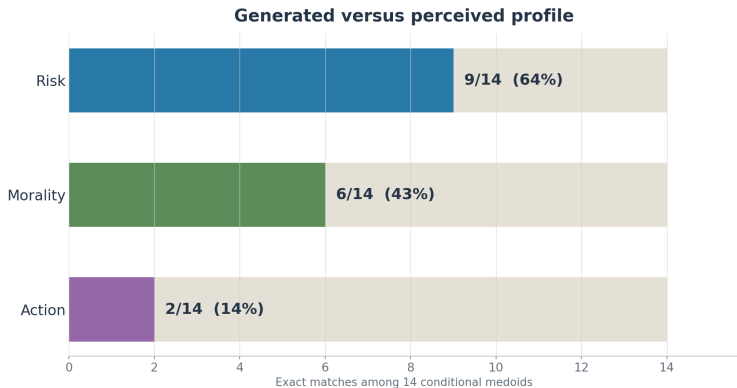
Individual trajectory

- Perceived profile: levels of risk, morality, and action.
- `profile_coherence`
- `causal_continuity`
- Choice references and short justifications

Pairwise comparison

- Relative profile shift on risk, morality, and action
- `narrative_distinctness`
- Evidence from stories A and B, with justification
- A/B and B/A orders used only to check order sensitivity.

Individual annotation results



Causal continuity	Profile coherence	Action recovery by level
14/14	9/14	Neutral 0/10; Physical 0/2; Tactical 2/2

Pairwise annotation results

Profile contrast	Outcome	Path difference (LCS)	A on Risk	A on Morality	A on Action	Narrative distinctness
Cautious (A) vs reckless (B)	Win	0.61	cautious	selfish	tactical	High
Cautious (A) vs reckless (B)	Death	0.79	cautious	selfish	tactical	High
Selfish (A) vs noble (B)	Win	0.41	cautious	selfish	tactical	Medium
Selfish (A) vs noble (B)	Death	0.21	cautious	selfish	tactical	Medium-high
Physical (A) vs tactical (B)	Win	0.38	even	noble	even	Medium-high
Physical (A) vs tactical (B)	Death	0.30	reckless	even	physical	Medium

Reading: bold = controlled axis; red = A/B-B/A disagreement averaged across orders, “main tendency” output displayed.

Intended profile contrast recovered in **5 of 6** comparisons.

Conclusion

Conclusion

What we learned

- Profiles reshape the probabilistic structure.
- Risk dominates the other axes.
- 5/6 relative story contrasts are recovered; absolute profiles remain unevenly visible.

Current limits

- One gamebook only.
- Simplified mechanics and state for the Markov model.
- Limited human calibration; no held-out validation.

Next

- Refine the LLM annotation pipeline.
- Compare the results with a more powerful LLM.
- Make a contrastive study on different books.
- Refine modeling of the game mechanics (e.g. combat)

This contribution is a **proof of concept for an auditable hybrid workflow** for analyzing gamebooks.

The LLM is used as a **controlled annotator, not a literary critic**.

BoP formalism provides **several indices on gamebooks**.

References



Aarseth, E. J. (1997).
Cybertext: perspectives on ergodic literature.
Johns Hopkins University Press, Baltimore, Md. London.



Dever, J. and Chalk, G. (1991).
Flight from the dark.
Red Fox, London.
OCLC: 226242649.



Françoise, K., Kivimäki, I., Mantrach, A., Rossi, F., and Saerens, M. (2017).
A bag-of-paths framework for network data analysis.
Neural Networks, 90:90–111.



Guex, G., Courtain, S., and Saerens, M. (2021).
Covariance and correlation measures on a graph in a generalized bag-of-paths formalism.
Journal of Complex Networks, 8(6):cnaa026.



Guex, G., Kivimäki, I., and Saerens, M. (2019).
Randomized optimal transport on a graph: framework and new distance measures.
Network Science, 7(1):88–122.



Montfort, N. (2005).
Twisty little passages: an approach to interactive fiction.
MIT Press, Cambridge, Mass. London.



Métrailler, C. E. (2025).
Une étude plurielle des réseaux de personnages : narrations interactives, jeux vidéo et adaptations.
PhD thesis.



Riedl, M. O. and Bulitko, V. (2013).
Interactive Narrative: An Intelligent Systems Approach.
AI Magazine, 34(1):67–77.



Wang, D. Y.-B., Shen, Z., Mishra, S. S., Xu, Z., Teng, Y., and Ding, H. (2025).
SLOT: Structuring the Output of Large Language Models.
Version Number: 1.

**Thank you for your
attention!**